

Guangzhi Chen

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EDUCATION

The University of Texas at Austin

August 2024 - December 2027

Bachelor of Science, Aerospace Engineering

Cumulative GPA: 3.91/4.00

Relevant Courses: Linear System Analysis, Feedback Control System, Spacecraft Dynamics, Flight Dynamics, Spacecraft Systems Design, Senior Design

EXPERIENCE

Software/GNC Engineer (Co-founder), **Texas Ascent & Landing**

July 2026 - Present

- Co-founded a student team developing an electric propulsive-landing hopper; building a reusable vehicle simulation in NASA's Trick framework with JEOD, structured as a general core with vehicle-specific configurations
- Unit-testing 6-DOF rigid-body dynamics and writing shell-script tests that check the sim against physical invariants (energy and momentum balance) & performance benchmarks, as groundwork for a CI/CD pipeline
- Implementing Monte Carlo dispersion runs within the team's Trick-based software layout, with automated plotting of the results

GNC Research Intern, **CDUS Lab (Dr. Maruthi Akella) - USSF SSTI**

June 2026 - July 2026

- Cut spacecraft position RMSE $7\times$ ($0.132\text{ m} \rightarrow 0.020\text{ m}$, below the 0.138 m measurement noise floor) by rebuilding a 15-state pose-estimation Multiplicative EKF around a 6-state Relative Orbital Element translation model, replacing a gravity-blind random walk with an exact Keplerian state-transition matrix
- Validated in Python on Stanford SLAB's SHIRT dataset (2,371 frames, 2 orbits of J2-perturbed truth) against the lab's existing baseline filter
- Derived the ROE-to-camera measurement map and its 3×6 Jacobian, and resolved the 3-measurement/6-state rank deficiency at initialization with a diffuse-prior, minimum-norm seed
- Cut maneuver-node innovation $45\times$ ($2.12\text{ m} \rightarrow 0.047\text{ m}$) by deriving a numeric ROE control-input matrix that injects known chaser impulse burns into the EKF predict step

Activities

Controls Engineer, **Quadcopter Controls Simulation (Personal Project)**

August 2025 - May 2026

- Developed a 6-DOF Quadcopter Simulator in C++, utilizing Eigen3 for rigid-body dynamics and implementing Runge-Kutta 4 (RK4)
- Implemented a safety-first motor mixing strategy that resolves actuator saturation by prioritizing thrust over attitude commands, preventing fly-aways and loss of altitude control
- Built a thrust-vector flight controller using cross-product-based attitude error rather than Euler angles, avoiding gimbal lock and enabling stable high-angle flight through a PID controller

Officer, **LUNAR Undergraduate Council, UT Aerospace**

November 2024 - Present

- Re-established the department's undergraduate advisory council post-COVID and launched the inaugural Student-Faculty Research Mixer, coordinating faculty to connect undergraduates with research opportunities
- Helped launch an alumni mentorship program (integrated with Cockrell's CAMP), recruiting 10+ senior aerospace/defense alumni (including NASA and Lockheed Orion GNC engineers) as mentors.

SKILLS & ACCOMPLISHMENTS

- **Technical Skills:** Python, C++, MATLAB, ROS 2, Gazebo, OOP, GitHub, SolidWorks, FEA, Soldering
- **Languages:** Chinese - fluent
- **Volunteering:** GivePulse, Storehouse of Collin County, Kids Against Hunger, ABC English Learning